

POCKET WOC EDITION

Source Code Transcription

Printable transcription of the treatment-advisor TypeScript and TSX source files.

Prepared from the workspace source files
File sections: 13

01 app/layout.tsx

```
import type { Metadata } from "next";
import { Geist, Geist_Mono } from "next/font/google";
import "./globals.css";

const geistSans = Geist({
  variable: "--font-geist-sans",
  subsets: ["latin"],
});

const geistMono = Geist_Mono({
  variable: "--font-geist-mono",
  subsets: ["latin"],
});

export const metadata: Metadata = {
  title: "Skintegrity Partners | Dressing Advisor",
  description: "Clinical Tool For Acute Care, Dressing Selection and Management",
};

export default function RootLayout({ children }: LayoutProps<"/">) {
  return (
    <html
      lang="en"
      className={` ${geistSans.variable} ${geistMono.variable} h-full antialiased` }
    >
      <body className="min-h-full flex flex-col">{children}</body>
    </html>
  );
}
```

02 app/page.tsx

```
'use client';

import React, { useState } from 'react';
import { initialWoundState, NavigationModule, WoundState } from '../types/wound';
import { Navbar } from './components/Navbar';
import { RoleSelection } from './components/RoleSelection';
import { LocationSelector } from './components/LocationSelector';
import { TriageEngine } from './components/TriageEngine';
import { CarePlanOutput } from './components/CarePlanOutput';
import {
  FormularyView,
  DebridementView,
  GlossaryView,
  CaseStudiesView,
} from './components/ModuleViews';

export default function Home() {
  const [woundState, setWoundState] = useState<WoundState>(initialWoundState);

  const updateState = (updates: Partial<WoundState>) => {
    setWoundState((prevWoundState: WoundState) => ({ ...prevWoundState, ...updates }));
  };

  const handleNextStep = () => {
    setWoundState((prevWoundState: WoundState) => ({
      ...prevWoundState,
      currentStep: prevWoundState.currentStep + 1,
    }));
  };

  const handleBackStep = () => {
    setWoundState((prevWoundState: WoundState) => ({
      ...prevWoundState,
      currentStep: Math.max(1, prevWoundState.currentStep - 1),
    }));
  };

  const handleReset = () => {
    setWoundState(initialWoundState);
  };

  const setActiveModule = (activeModule: NavigationModule) => {
    setWoundState((prevWoundState: WoundState) => ({ ...prevWoundState, activeModule }));
  };

  return (
    <div className="min-h-screen bg-slate-100 flex flex-col font-sans">
      {/* Top Navbar */}
      <Navbar
        activeModule={woundState.activeModule}
        setActiveModule={setActiveModule}
        role={woundState.role}
        resetAssessment={handleReset}
      />

      {/* Main Flow & Module Controller */}
      <main className="flex-1 pb-12">
        {woundState.activeModule === 'ASSESSMENT' && (
          <>
            {woundState.currentStep === 1 && (
              <RoleSelection
                state={woundState}
                updateState={updateState}
                onNext={handleNextStep}
              />
            )}

            {woundState.currentStep === 2 && (
              <LocationSelector
                state={woundState}
                updateState={updateState}
                onNext={handleNextStep}
                onBack={handleBackStep}
              />
            )}
          </>
        )}
      </main>
    </div>
  );
}
```

```

        {woundState.currentStep === 3 && (
            <TriageEngine
                state={woundState}
                updateState={updateState}
                onNext={handleNextStep}
                onBack={handleBackStep}
            />
        )}

        {woundState.currentStep === 4 && (
            <CarePlanOutput
                state={woundState}
                onReset={handleReset}
                onBack={handleBackStep}
            />
        )}
    </>
)}

{/* Supplemental Navigation Views */}
{woundState.activeModule === 'FORMULARY' && <FormularyView />}
{woundState.activeModule === 'DEBRIDEMENT' && <DebridementView />}
{woundState.activeModule === 'GLOSSARY' && <GlossaryView />}
{woundState.activeModule === 'CASES' && <CaseStudiesView />}
</main>
</div>
);
}

```

03 types/wound.ts

```
export type UserRole = 'RN' | 'PROVIDER' | null;

export type NavigationModule = 'ASSESSMENT' | 'FORMULARY' | 'DEBRIDEMENT' | 'GLOSSARY' | 'CASES';

export type AnatomicalZone =
  | 'Sacrum' | 'Coccyx' | 'Heels' | 'Ischial Tuberosities'
  | 'Trochanter' | 'Occiput' | 'Malleolus' | 'Foot/Toes'
  | 'Forearm' | 'Abdomen' | 'Other';

export type WoundEtiology =
  | 'PRESSURE_INJURY'
  | 'SKIN_TEAR'
  | 'MASD_IAD'
  | 'VENOUS_ULCER'
  | 'ARTERIAL_ULCER'
  | 'DIABETIC_FOOT_ULCER'
  | 'SURGICAL_DEHISCENCE'
  | 'UNKNOWN';

export interface WoundState {
  // Global & Flow
  role: UserRole;
  activeModule: NavigationModule;
  currentStep: number;

  // Patient & Context
  paceMode: 'FAST' | 'DETAILED';
  bradenScore?: number;
  isIncontinent: boolean;

  // Location
  anatomicalSite: AnatomicalZone | null;
  isBonyProminence: boolean;
  isLowerExtremity: boolean;

  // Triage & Assessment
  etiology: WoundEtiology;
  isFullThickness: boolean;
  hasSloughOrEschar: boolean;
  hasTunnelingOrUndermining: boolean;
  isWorsened: boolean;
  isHAPI: boolean;
  stage2Count: number;

  // Mandatory Consult Flags (Auto-calculated)
  triggers: {
    wocRNConsult: boolean;
    podiatryVascular: boolean;
    surgicalEval: boolean;
    nutritionConsult: boolean;
  };
};

export const initialWoundState: WoundState = {
  role: null,
  activeModule: 'ASSESSMENT',
  currentStep: 1,
  paceMode: 'FAST',
  isIncontinent: false,
  anatomicalSite: null,
  isBonyProminence: false,
  isLowerExtremity: false,
  etiology: 'UNKNOWN',
  isFullThickness: false,
  hasSloughOrEschar: false,
  hasTunnelingOrUndermining: false,
  isWorsened: false,
  isHAPI: false,
  stage2Count: 0,
  triggers: {
    wocRNConsult: false,
    podiatryVascular: false,
    surgicalEval: false,
    nutritionConsult: false,
  }
};
```

},
};

04 app/components/LocationSelector.tsx

```
'use client';

import React from 'react';
import { WoundState } from '../../types/wound';
import InteractiveBody3D from './InteractiveBody3D';

interface LocationSelectorProps {
  state: WoundState;
  updateState: (updates: Partial<WoundState>) => void;
  onNext: () => void;
  onBack: () => void;
}

export const LocationSelector: React.FC<LocationSelectorProps> = ({
  state,
  updateState,
  onNext,
  onBack,
}) => {
  const InteractiveBody3DWithSelection = InteractiveBody3D as React.ComponentType<{
    selectedZone: string | null;
    onSelect?: (zone: string) => void;
  }>;

  const handleZoneSelect = (zone: string) => {
    const isLower = [
      'Heel (Calcaneus)',
      'Malleolus (Ankle)',
      'Foot / Toes',
      'Distal Lower Extremity',
    ].includes(zone);

    updateState({
      anatomicalSite: zone as any,
      isLowerExtremity: isLower,
    });
  };

  const locations = [
    { id: 'Sacrum', label: 'Sacrum', desc: 'High-risk pressure site over sacral bone', tag: 'BONY' },
    { id: 'Greater Trochanter', label: 'Greater Trochanter', desc: 'Hip prominence from side-lying pressure', tag: 'BONY' },
    { id: 'Ischial Tuberosity', label: 'Ischial Tuberosity', desc: 'Seated pressure site ("sit bones")', tag: 'BONY' },
    { id: 'Malleolus (Ankle)', label: 'Malleolus (Ankle)', desc: 'Inner/outer ankle prominence', tag: 'BONY', extraTag: 'LOWER EXTREMITY' },
  ],
  { id: 'Heel (Calcaneus)', label: 'Heel (Calcaneus)', desc: 'Triggers strict foot offloading logic', tag: 'BONY', extraTag: 'LOWER EXTREMITY' },
  { id: 'Foot / Toes', label: 'Foot / Toes', desc: 'Dorsum, plantar, or interdigital lesions', tag: 'BONY', extraTag: 'LOWER EXTREMITY' },
];

  return (
    <div className="max-w-6xl mx-auto py-8 px-4 sm:px-6">
      <div className="flex justify-between items-center mb-6">
        <div>
          <h2 className="text-2xl font-bold text-slate-900">Anatomical Location & X-Ray View</h2>
          <p className="text-sm text-slate-600">Select a site to auto-zoom and inspect underlying bony prominences.</p>
        </div>
        <span className="text-xs font-semibold text-slate-500 bg-slate-200 px-3 py-1 rounded-full">
          Step 2 of 4
        </span>
      </div>

      <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
        <div className="lg:col-span-6">
          <div className="lg:col-span-6">
            <InteractiveBody3DWithSelection
              selectedZone={state.anatomicalSite ?? null}
              onSelect={handleZoneSelect}
            />
          </div>

          <div>
            <div>
              <div>
                <div>
                  <div>
                    <div>
                      <div>
                        <div>
                          <div>
                        </div>
                      </div>
                    </div>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </div>

        <div>
          <div>
            <div>
              <div>
                <div>
                  <div>
                    <div>
                      <div>
                        <div>
                          <div>
                        </div>
                      </div>
                    </div>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  );
}
```

```

const isSelected = state.anatomicalSite === loc.id;
return (
  <button
    key={loc.id}
    onClick={() => handleZoneSelect(loc.id)}
    className={`p-4 rounded-xl border text-left transition-all flex flex-col justify-between ${
      isSelected
        ? 'border-blue-600 bg-blue-50 ring-2 ring-blue-500/20 shadow-sm'
        : 'border-slate-200 bg-white hover:border-slate-300'
      }`}
  >
    <div>
      <div className="flex items-center justify-between">
        <span className="text-sm font-bold text-slate-900">{loc.label}</span>
        {loc.tag && (
          <span className="text-[10px] font-extrabold bg-amber-100 text-amber-800 px-1.5 py-0.5 rounded">
            {loc.tag}
          </span>
        )}
      </div>
      <p className="text-xs text-slate-500 mt-1">{loc.desc}</p>
    </div>
    {loc.extraTag && (
      <span className="mt-2 text-[9px] font-bold bg-purple-100 text-purple-800 px-1.5 py-0.5 rounded self-start">
        {loc.extraTag}
      </span>
    )}
  </button>
);
)}}
</div>

<div className="pt-4 border-t border-slate-200 flex justify-between items-center">
  <button onClick={onBack} className="px-4 py-2 text-xs font-bold text-slate-600 hover:text-slate-900">
    ← Back
  </button>
  <button
    onClick={onNext}
    disabled={!state.anatomicalSite}
    className={`px-6 py-2.5 rounded-lg text-xs font-bold text-white transition-all ${
      state.anatomicalSite ? 'bg-blue-600 hover:bg-blue-700 shadow-md' : 'bg-slate-300 cursor-not-allowed'
    }`}
  >
    Continue to Triage Engine →
  </button>
</div>
</div>
</div>
</div>
);
};

```

```

'use client';

import React from 'react';
type WoundEtiology =
  | 'PRESSURE_INJURY'
  | 'SKIN_TEAR'
  | 'MASD_IAD'
  | 'VENOUS_ULCER'
  | 'ARTERIAL_ULCER'
  | 'DIABETIC_FOOT_ULCER'
  | 'SURGICAL_DEHISCENCE'
  | 'UNKNOWN';

interface WoundTriggers {
  wocRNConsult: boolean;
  podiatryVascular: boolean;
  surgicalEval: boolean;
  nutritionConsult: boolean;
}

interface WoundState {
  etiology: WoundEtiology;
  isFullThickness: boolean;
  hasSloughOrEschar: boolean;
  hasTunnelingOrUndermining: boolean;
  isHAPI: boolean;
  isWorsened: boolean;
  isLowerExtremity: boolean;
  stage2Count: number;
  triggers: WoundTriggers;
}

interface TriageEngineProps {
  state: WoundState;
  updateState: (updates: Partial<WoundState>) => void;
  onNext: () => void;
  onBack: () => void;
}

export const TriageEngine: React.FC<TriageEngineProps> = ({
  state,
  updateState,
  onNext,
  onBack,
}) => {
  // Recalculate triggers whenever assessment attributes change
  const handleAttributeChange = (updates: Partial<WoundState>) => {
    const newState = { ...state, ...updates };

    // Trigger Logic Rules
    const isFullThickness = newState.isFullThickness || newState.hasSloughOrEschar;
    const isStage3Plus = isFullThickness; // Slough or full thickness loss maps to Stage 3+ / Unstageable equivalent

    // Nutrition Trigger Rule: Stage 3+, Full Thickness, DTPI, or MULTIPLE Stage 2s
    const nutritionConsult =
      isFullThickness ||
      newState.stage2Count > 1;

    // WOC-RN Trigger Rule: Stage 3+, HAPI, Full Thickness, or Worsened
    const wocRNConsult =
      isStage3Plus ||
      newState.isHAPI ||
      newState.isWorsened;

    // Podiatry / Vascular Trigger Rule: Full thickness / Stage 3+ on lower extremity
    const podiatryVascular =
      newState.isLowerExtremity && (isFullThickness || isStage3Plus);

    // Surgical Debridement Trigger Rule: Presence of slough or eschar
    const surgicalEval = newState.hasSloughOrEschar;

    updateState({
      ...updates,
      isFullThickness,
      triggers: {

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```

        wocRNConsult,
        podiatryVascular,
        surgicalEval,
        nutritionConsult,
    },
  });
});

const etiologies: { id: WoundEtiology; label: string }[] = [
  { id: 'PRESSURE_INJURY', label: 'Pressure Injury (PI / DTPI)' },
  { id: 'SKIN_TEAR', label: 'Skin Tear / Traumatic Mechanical' },
  { id: 'MASD_IAD', label: 'Incontinence Dermatitis (IAD / MASD)' },
  { id: 'VENOUS_ULCER', label: 'Venous Stasis Ulcer' },
  { id: 'ARTERIAL_ULCER', label: 'Arterial Ulcer' },
  { id: 'DIABETIC_FOOT_ULCER', label: 'Diabetic / Neuropathic Foot Ulcer' },
  { id: 'SURGICAL_DEHISCENCE', label: 'Surgical Dehiscence' },
  { id: 'UNKNOWN', label: 'Suspected / Unknown Etiology' },
];

return (
  <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
    { /* Header */ }
    <div className="mb-6 flex items-center justify-between">
      <div>
        <h2 className="text-2xl font-bold text-slate-900">Physical Characteristics & Depth</h2>
        <p className="text-sm text-slate-600">Select observed tissue layers and clinical indicators.</p>
      </div>
      <span className="text-xs font-semibold px-3 py-1 bg-slate-100 text-slate-700 rounded-full">
        Step 3 of 4
      </span>
    </div>

    { /* Etiology Selection */ }
    <div className="mb-8">
      <label className="block text-xs font-bold text-slate-700 uppercase tracking-wide mb-3">
        1. Primary Suspected Etiology
      </label>
      <div className="grid grid-cols-2 md:grid-cols-4 gap-2">
        {etiologies.map((item) => (
          <button
            key={item.id}
            type="button"
            onClick={() => handleAttributeChange({ etiology: item.id })}
            className={`p-3 rounded-md border text-xs font-semibold text-left transition-all ${
              state.etiology === item.id
                ? 'border-blue-600 bg-blue-50 text-blue-900 ring-1 ring-blue-600'
                : 'border-slate-200 bg-white text-slate-700 hover:bg-slate-50'
            }`}
          >
            {item.label}
          </button>
        ))}
      </div>
    </div>

    { /* Characteristic Checkboxes */ }
    <div className="mb-8 space-y-4">
      <label className="block text-xs font-bold text-slate-700 uppercase tracking-wide">
        2. Tissue Assessment & Cues
      </label>

      { /* Slough / Eschar */ }
      <div>
        onClick={() => handleAttributeChange({ hasSloughOrEschar: !state.hasSloughOrEschar })}
        className={`p-4 rounded-lg border cursor-pointer transition-all ${
          state.hasSloughOrEschar ? 'border-amber-500 bg-amber-50/40' : 'border-slate-200 bg-white'
        }`}
      >
        <div className="flex items-start gap-3">
          <input
            type="checkbox"
            checked={state.hasSloughOrEschar}
            onChange={() => {}} // Handled by div click
            className="mt-1 h-4 w-4 rounded border-slate-300 text-blue-600"
          />
          <div>
            <span className="text-sm font-bold text-slate-900">Devitalized Tissue Present (Slough or Eschar)</span>
            <p className="text-xs text-slate-600">
              Yellow, tan, stringy slough or hard black/brown eschar noted. <i>Automatically classifies wound as Full-Thickness tissue

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loss.</i>
    </p>
  </div>
</div>
</div>

{/* Depth / Full-Thickness */}
<div
  onClick={() => handleAttributeChange({ isFullThickness: !state.isFullThickness })}
  className={`p-4 rounded-lg border cursor-pointer transition-all ${
    state.isFullThickness ? 'border-blue-600 bg-blue-50/40' : 'border-slate-200 bg-white'
  }`}
>
  <div className="flex items-start gap-3">
    <input
      type="checkbox"
      checked={state.isFullThickness}
      onChange={() => {}}
      className="mt-1 h-4 w-4 rounded border-slate-300 text-blue-600"
    />
    <div>
      <span className="text-sm font-bold text-slate-900">Visible Subcutaneous Fat, Muscle, Tendon, or Bone</span>
      <p className="text-xs text-slate-600">
        Depth extends beyond the dermal layer into underlying structures (Full-Thickness Loss / Stage 3+).
      </p>
    </div>
  </div>
</div>

{/* Tunneling / Undermining */}
<div
  onClick={() => handleAttributeChange({ hasTunnelingOrUndermining: !state.hasTunnelingOrUndermining })}
  className={`p-4 rounded-lg border cursor-pointer transition-all ${
    state.hasTunnelingOrUndermining ? 'border-blue-600 bg-blue-50/40' : 'border-slate-200 bg-white'
  }`}
>
  <div className="flex items-start gap-3">
    <input
      type="checkbox"
      checked={state.hasTunnelingOrUndermining}
      onChange={() => {}}
      className="mt-1 h-4 w-4 rounded border-slate-300 text-blue-600"
    />
    <div>
      <span className="text-sm font-bold text-slate-900">Tunneling or Undermining Present</span>
      <p className="text-xs text-slate-600">
        Narrow channels extending from wound bed or tissue shelf destruction under intact skin margins. Requires gelling fiber
        packing.
      </p>
    </div>
  </div>
</div>

{/* Worsened Status */}
<div
  onClick={() => handleAttributeChange({ isWorsened: !state.isWorsened })}
  className={`p-4 rounded-lg border cursor-pointer transition-all ${
    state.isWorsened ? 'border-red-500 bg-red-50/40' : 'border-slate-200 bg-white'
  }`}
>
  <div className="flex items-start gap-3">
    <input
      type="checkbox"
      checked={state.isWorsened}
      onChange={() => {}}
      className="mt-1 h-4 w-4 rounded border-slate-300 text-blue-600"
    />
    <div>
      <span className="text-sm font-bold text-slate-900">Wound Has WORSENERD Since Prior Assessment</span>
      <p className="text-xs text-slate-600">
        Increase in size, depth, devitalized tissue load, or exudate. Triggers mandatory WOC-RN escalation.
      </p>
    </div>
  </div>
</div>
</div>

{/* Multiple Stage 2 Counter */}
<div className="p-4 bg-slate-50 border border-slate-200 rounded-lg mb-8">

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<div className="flex items-center justify-between">
  <div>
    <span className="text-sm font-bold text-slate-900">Multiple Stage 2 Pressure Injuries?</span>
    <p className="text-xs text-slate-500">Count of partial-thickness pressure injuries across all sites.</p>
  </div>
  <div className="flex items-center gap-3">
    <button
      type="button"
      onClick={() => handleAttributeChange({ stage2Count: Math.max(0, state.stage2Count - 1) })}
      className="w-8 h-8 rounded-md bg-white border border-slate-300 font-bold text-slate-700 hover:bg-slate-100"
    >
      -
    </button>
    <span className="text-sm font-bold text-slate-900 w-4 text-center">{state.stage2Count}</span>
    <button
      type="button"
      onClick={() => handleAttributeChange({ stage2Count: state.stage2Count + 1 })}
      className="w-8 h-8 rounded-md bg-white border border-slate-300 font-bold text-slate-700 hover:bg-slate-100"
    >
      +
    </button>
  </div>
</div>
</div>

{/* Active Consult Triggers Indicator */}
<div className="p-4 bg-slate-900 text-white rounded-lg mb-8">
  <span className="block text-xs font-bold text-slate-400 uppercase tracking-wide mb-2">
    Real-Time Consultation Triggers Active:
  </span>
  <div className="flex flex-wrap gap-2">
    {state.triggers.wocRNConsult && (
      <span className="px-2.5 py-1 rounded bg-blue-900 border border-blue-700 text-xs font-semibold text-blue-200">
        ✓ WOC-RN Consult
      </span>
    )}
    {state.triggers.podiatryVascular && (
      <span className="px-2.5 py-1 rounded bg-purple-900 border border-purple-700 text-xs font-semibold text-purple-200">
        ✓ Podiatry / Vascular (Primary)
      </span>
    )}
    {state.triggers.surgicalEval && (
      <span className="px-2.5 py-1 rounded bg-amber-900 border border-amber-700 text-xs font-semibold text-amber-200">
        ✓ Surgical Debridement Eval
      </span>
    )}
    {state.triggers.nutritionConsult && (
      <span className="px-2.5 py-1 rounded bg-emerald-900 border border-emerald-700 text-xs font-semibold text-emerald-200">
        ✓ Clinical Nutrition Consult
      </span>
    )}
    {!Object.values(state.triggers).some(Boolean) && (
      <span className="text-xs text-slate-400 italic">No specialty consults currently triggered.</span>
    )}
  </div>
</div>

{/* Navigation Controls */}
<div className="flex justify-between pt-4 border-t border-slate-200">
  <button
    type="button"
    onClick={onBack}
    className="px-5 py-2.5 rounded-lg text-sm font-medium border border-slate-300 text-slate-700 hover:bg-slate-50"
  >
    ← Back
  </button>
  <button
    type="button"
    onClick={onNext}
    className="px-6 py-2.5 rounded-lg text-sm font-semibold bg-blue-600 text-white hover:bg-blue-700 shadow-md"
  >
    Generate Care Plan & Output →
  </button>
</div>
</div>
);
};

```

```
// types/pressureInjury.ts

export type AnatomicalLocation =
  | 'Sacrum/Coccyx/Perineal'
  | 'Heel/Foot/Toe'
  | 'Trochanter/Ischium'
  | 'Other Bony Prominence'
  | 'Non-Bony Prominence';

export type WoundBedTissue =
  | 'Intact Discolored / DTPI'
  | 'Clean Granulating / Red'
  | 'Slough / Yellow ( < 50% )'
  | 'Slough / Yellow ( >= 50% )'
  | 'Eschar / Black ( < 50% )'
  | 'Eschar / Black ( >= 50% )';

export type Stage = 'Stage 1' | 'Stage 2' | 'Stage 3' | 'Stage 4' | 'Unstageable' | 'DTPI';

export type PressureInjuryInput = {
  stage: Stage;
  location: AnatomicalLocation;
  tissuePresentation: WoundBedTissue;
  hasMedicalDevice: boolean;
  exudate: 'None/Minimal' | 'Moderate' | 'Heavy';
  isIncontinenceAtRisk: boolean;
  isHighFrictionShear: boolean;
  hasCavityOrTunneling: boolean;
  hasInfectionSigns: boolean;
  isBloodyBullae: boolean;
  isArterialInsufficiency: boolean;
  isCrossHatchingConsidered: boolean;
};

export type RuleOutput = {
  category: string;
  cleanser?: string;
  primaryDressing: string;
  secondaryDressing?: string;
  changeFrequency: string;
  removalProtocol: string;
  preventionStandard: string;
  alerts: string[];
};

export function evaluatePressureInjury(input: PressureInjuryInput): RuleOutput {
  const alerts: string[] = [];

  // Base Prevention Standard appended to all outputs
  const preventionStandard = "Follow pressure injury prevention measures, including regular turning and offloading, per facility protocols.";
  const standardRemoval = "Apply adhesive remover spray/wipes around border. Push skin down gently while pulling dressing back parallel to skin.";

  // -----
  // 1. HARD-STOP OVERRIDES & CONSULT TRIGGERS
  // -----

  // Non-Bony Prominence Override
  if (input.location === 'Non-Bony Prominence' && !input.hasMedicalDevice) {
    alerts.push("CRITICAL ETIOLOGY CHECK: Location is over a non-bony prominence without an associated medical device. This presentation is uncharacteristic of a Pressure Injury. Please confer with a WOC-RN / Wound Specialist to re-evaluate etiology (e.g., MASD, Vascular, Traumatic).");
  }

  // Arterial Insufficiency Foot/Toe Hard-Stop
  if (input.location === 'Heel/Foot/Toe' && input.isArterialInsufficiency) {
    return {
      category: "SURGICAL CONSULT REQUIRED – ARTERIAL ISCHEMIA",
      primaryDressing: "Keep wound bed dry and clean. Do NOT debride or apply occlusive/moisture-retentive dressings.",
      changeFrequency: "Inspect daily.",
      removalProtocol: "N/A – Keep open to air or apply dry protective dressing.",
      preventionStandard,
      alerts: [
        ...alerts,

```

```

        "HARD STOP: Suspected/known arterial insufficiency on foot/toe. Consult appropriate surgical service immediately (Podiatry vs.
Vascular Surgery)."
```

```

    ]
  };
}

// Major Devitalized Tissue (>= 50%) Surgical Trigger
if (input.tissuePresentation === 'Slough / Yellow ( >= 50% )' || input.tissuePresentation === 'Eschar / Black ( >= 50% )') {
  alerts.push("SURGICAL CONSULT RECOMMENDED: 50% or more of the wound base is covered by devitalized tissue. Consider surgical consult
for formal debridement evaluation.");
}

// Cross-Hatching Trigger
if (input.isCrossHatchingConsidered) {
  alerts.push("SURGICAL CONSULT REQUIRED: Cross-hatching eschar requires specialized surgical/WOC-RN consult. Do not attempt bedside
cross-hatching without authorization.");
}

// -----
// 2. STAGE-SPECIFIC DECISION LOGIC
// -----

// --- STAGE 1 ---
if (input.stage === 'Stage 1') {
  if (input.isIncontinenceAtRisk && input.isHighFrictionShear) {
    return {
      category: "Prophylactic & Moisture Protection",
      primaryDressing: "No-Sting Barrier Film (allow to dry completely), then apply a 5-layer bordered soft silicone foam dressing.",
      secondaryDressing: input.tissuePresentation === 'Intact Discolored / DTPI' ? "If skin appears fragile, apply a non-adherent
contact layer prior to foam." : undefined,
      changeFrequency: "Inspect skin/dressing daily; change QOD or when dislodged/soiled.",
      removalProtocol: standardRemoval,
      preventionStandard,
      alerts
    };
  }
  if (input.isIncontinenceAtRisk) {
    return {
      category: "Incontinence Moisture Barrier",
      primaryDressing: "Apply moisture barrier emollient cream/ointment.",
      changeFrequency: "Apply twice daily (BID) and after each episode of incontinence.",
      removalProtocol: "Gently cleanse barrier cream with pH-balanced cleanser; do not scrub skin.",
      preventionStandard,
      alerts
    };
  }
  return {
    category: "Protective Barrier",
    primaryDressing: "No-Sting Barrier Film (allow to dry completely), then cover with Hydrocolloid Dressing.",
    changeFrequency: "Change Q3-7 days or if compromised.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}

// --- STAGE 2 ---
if (input.stage === 'Stage 2') {
  if (input.isBloodyBullae && input.location !== 'Sacrum/Coccyx/Perineal') {
    return {
      category: "Bloody Bullae Protection",
      primaryDressing: "Hydrocolloid Dressing",
      changeFrequency: "Change Q3-7 days or when fluid reaches margin.",
      removalProtocol: standardRemoval,
      preventionStandard,
      alerts
    };
  }
  if (input.exudate === 'Heavy') {
    return {
      category: "Absorbent Gelling Strategy",
      primaryDressing: "Calcium Alginate Pad or Gelling Hydrofiber Sheet sized strictly within wound margins.",
      secondaryDressing: "Cover with 5-Layer Soft Silicone Foam Dressing or absorbent pad.",
      changeFrequency: "Change Q1-3 days and PRN for strike-through drainage.",
      removalProtocol: `${standardRemoval} Rehydrate alginate/hydrofiber with normal saline if adherent before removal.`,
      preventionStandard,
      alerts
    };
  }
}

```

```

if (input.exudate === 'Moderate') {
  return {
    category: "Exudate Management & Friction Protection",
    primaryDressing: "No-Sting Barrier Film to peri-wound (let dry). Apply 5-Layer Bordered Soft Silicone Foam Dressing.",
    secondaryDressing: "If peri-wound skin is fragile, place a non-adherent silicone contact layer directly over wound bed first.",
    changeFrequency: "Inspect daily; change Q3D or when exudate reaches 1-2 cm from border.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}
// Low / Minimal Exudate
return {
  category: "Moist Healing & Non-Adherent Contact",
  primaryDressing: "No-Sting Barrier Film to peri-wound (let dry). Apply Non-Adherent Contact Layer directly to wound bed.",
  secondaryDressing: "Dry island dressing or thin foam cover.",
  changeFrequency: "Change Q3-7 days.",
  removalProtocol: standardRemoval,
  preventionStandard,
  alerts
};
}

// --- STAGE 3 & STAGE 4 ---
if (input.stage === 'Stage 3' || input.stage === 'Stage 4') {
  // Infection Sub-pathway
  if (input.hasInfectionSigns) {
    return {
      category: "Antimicrobial & Infection Control",
      cleanser: "Irrigate with Antimicrobial Wound Cleanser or Sodium Hypochlorite (Dakin's 0.25% / quarter-strength) solution.",
      primaryDressing: input.exudate === 'Heavy' || input.exudate === 'Moderate'
        ? "Medical-Grade Honey Impregnated Sheet/Rope or Silver Alginate/Hydrofiber Rope."
        : "Medical-Grade Honey Gel or Silver Hydrogel (apply nickel-thick) packed with saline-dampened gauze.",
      secondaryDressing: "Bordered Multi-Layered Soft Silicone Foam Dressing.",
      changeFrequency: "Change QD to QOD, and PRN for strike-through drainage. Re-evaluate infection in 72 hours.",
      removalProtocol: standardRemoval,
      preventionStandard,
      alerts
    };
  }
}

// Enzymatic Debridement Sub-pathway (Non-infected with slough/eschar)
if (input.tissuePresentation.includes('Slough') || input.tissuePresentation.includes('Eschar')) {
  return {
    category: "Enzymatic Debridement Strategy",
    cleanser: "Irrigate with sterile normal saline (avoid acidic or heavy metal cleansers that denature enzymes).",
    primaryDressing: "Apply Collagenase Ointment (Santyl) nickel-thick across entire wound bed (both viable and non-viable tissue).",
    secondaryDressing: "Pack gently with saline-dampened gauze (wet-to-moist) and cover with non-adherent secondary foam.",
    changeFrequency: "Change Daily (QD).",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}

// Deep Cavity / Tunneling Management
if (input.hasCavityOrTunneling) {
  if (input.exudate === 'Heavy' || input.exudate === 'Moderate') {
    return {
      category: "Cavity Packing & Exudate Absorption",
      primaryDressing: "Fill cavity loosely with Calcium Alginate Rope/Ribbon or Gelling Hydrofiber Rope.",
      secondaryDressing: "Bordered Multi-Layered Soft Silicone Foam Dressing.",
      changeFrequency: "Change Q1-3 days and PRN for strike-through drainage.",
      removalProtocol: `${standardRemoval} Rehydrate gelling rope prior to extraction if adherent.`,
      preventionStandard,
      alerts
    };
  }
  return {
    category: "Wet-to-Moist Cavity Packing",
    primaryDressing: "Apply nickel-thick layer of wound gel to cavity base. Pack cavity gently with saline-dampened packing gauze (1-2 layers laid flat).",
    secondaryDressing: "Bordered Multi-Layered Soft Silicone Foam Dressing.",
    changeFrequency: "Change Daily (QD) and PRN for strike-through drainage.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}
}

```

```

// Shallow Full-Thickness Management
if (input.exudate === 'Heavy') {
  return {
    category: "Full-Thickness Heavy Exudate Management",
    primaryDressing: "Calcium Alginate Pad or Gelling Hydrofiber Sheet sized strictly within wound margins.",
    secondaryDressing: "Bordered Multi-Layered Soft Silicone Foam Dressing.",
    changeFrequency: "Change Q1-3 days and PRN for strike-through drainage.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}
return {
  category: "Wet-to-Moist Shallow Wound Bed Strategy",
  primaryDressing: "Apply nickel-thick layer of wound gel to wound base. If depth allows, pack gently with saline-dampened gauze (1-2
layers laid flat).",
  secondaryDressing: "Bordered Multi-Layered Soft Silicone Foam Dressing.",
  changeFrequency: "Change Daily (QD) to Q3D, and PRN for strike-through drainage.",
  removalProtocol: standardRemoval,
  preventionStandard,
  alerts
};
}

// --- UNSTAGEABLE ---
if (input.stage === 'Unstageable') {
  // Stable Heel Eschar Guardrail
  if (input.location === 'Heel/Foot/Toe' && input.tissuePresentation.includes('Eschar') && !input.hasInfectionSigns) {
    return {
      category: "Protective Eschar Offloading Strategy",
      primaryDressing: "Paint with Povidone-Iodine (Betadine) swab or leave open to air.",
      secondaryDressing: "Offload completely using heel elevation device/boot.",
      changeFrequency: "Inspect skin daily for softening, drainage, or peri-wound erythema.",
      removalProtocol: "N/A – Keep dry.",
      preventionStandard,
      alerts: [
        ...alerts,
        "CLINICAL CAUTION: Do not debride, soften, or apply moist dressings to stable, dry, intact eschar on heels."
      ]
    };
  }
}

// Infected Unstageable Debridement
if (input.hasInfectionSigns) {
  return {
    category: "Infected Unstageable Debridement Strategy",
    cleanser: "Irrigate thoroughly with Antimicrobial Wound Cleanser or Sodium Hypochlorite (Dakin's 0.25% / quarter-strength).",
    primaryDressing: "Apply Medical-Grade Honey Ointment/Gel across wound bed. Pack loosely with saline-dampened gauze or apply
honey-impregnated dressing.",
    secondaryDressing: "Cover with secondary soft silicone foam.",
    changeFrequency: "Change Daily (QD) and PRN for strike-through drainage.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
  };
}

// Clean Unstageable Debridement (Santyl)
return {
  category: "Enzymatic Debridement (Santyl Strategy)",
  primaryDressing: "Apply Collagenase Ointment (Santyl) nickel-thick across entire wound bed (both viable and non-viable tissue).",
  secondaryDressing: "Cover with saline-dampened gauze (wet-to-moist) and non-adherent silicone foam cover.",
  changeFrequency: "Change Daily (QD).",
  removalProtocol: standardRemoval,
  preventionStandard,
  alerts
};
}

// --- DEEP TISSUE PRESSURE INJURY (DTPI) ---
if (input.stage === 'DTPI') {
  // DTPI Foot/Toe Guardrail: Do not cover with foam
  if (input.location === 'Heel/Foot/Toe') {
    return {
      category: "Lower Extremity DTPI Protection Strategy",
      primaryDressing: "Apply Non-Sting Barrier Film to intact skin. Apply Non-Adherent Contact Layer.",
      secondaryDressing: "Light protective gauze wrapping or offloading boot. DO NOT use occlusive foam dressings.",
      changeFrequency: "Inspect skin daily; reapply barrier film as needed.",
    };
  }
}

```

```

        removalProtocol: standardRemoval,
        preventionStandard,
        alerts: [
            ...alerts,
            "CLINICAL GUARDRAIL: Do not cover DTPIs over the foot/heel with foam dressings as this promotes unwanted autolytic breakdown
and maceration."
        ]
    };
}

// Flat & Non-Raised DTPI
return {
    category: "Non-Raised DTPI Protection Strategy",
    primaryDressing: "Apply Non-Sting Barrier Film (allow to dry). Place Non-Adherent Contact Layer over intact discolored area.",
    secondaryDressing: "5-Layer Bordered Multi-Layered Soft Silicone Foam Dressing.",
    changeFrequency: "Inspect skin under dressing daily; change QOD or when soiled.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
};
}

// Default fallback
return {
    category: "General Protection Strategy",
    primaryDressing: "Non-Sting Barrier Film and Non-Adherent Silicone Contact Layer.",
    secondaryDressing: "Bordered Soft Silicone Foam Dressing.",
    changeFrequency: "Change Q3D or PRN.",
    removalProtocol: standardRemoval,
    preventionStandard,
    alerts
};
}

```

07 app/components/CarePlanOutput.tsx

```
'use client';

import React, { useState } from 'react';
import type { WoundState } from '../../types/wound';

interface CarePlanOutputProps {
  state: WoundState;
  onReset: () => void;
  onBack: () => void;
}

export const CarePlanOutput: React.FC<CarePlanOutputProps> = ({
  state,
  onReset,
  onBack,
}) => {
  const [copied, setCopied] = useState(false);

  // Generate dynamic clinical narrative
  const generateNote = () => {
    const isFootOrAnkle = state.isLowerExtremity;
    const isFullThickness = state.isFullThickness || state.hasSloughOrEschar;

    let text = `=====\\n`;
    text += `          POCKET WOC - CLINICAL ASSESSMENT & CARE PLAN          \\n`;
    text += `=====\\n\\n`;

    text += `ASSESSMENT SUMMARY:\\n`;
    text += `- Clinical Role: ${state.role === 'RN' ? 'Bedside Registered Nurse' : 'Provider (MD/NP/PA/DPM)'}\\n`;
    text += `- Lesion Location: ${state.anatomicalSite || 'Not specified'} (${state.isBonyProminence ? 'Bony Prominence' : 'Soft Tissue'})\\n`;
    text += `- Primary Etiology: ${state.etiology.replace('_', ' ')}\\n`;
    text += `- Tissue Depth: ${isFullThickness ? 'FULL-THICKNESS Loss (Subcutaneous/Fascia/Muscle/Bone)' : 'PARTIAL-THICKNESS Loss (Epidermis/Dermis)'}\\n`;
    text += `- Devitalized Tissue (Slough/Eschar): ${state.hasSloughOrEschar ? 'YES (Present)' : 'None noted'}\\n`;
    text += `- Tunneling / Undermining: ${state.hasTunnelingOrUndermining ? 'YES (Present)' : 'None'}\\n`;
    text += `- Assessment Status: ${state.isWorsened ? 'WORSENER compared to prior evaluation' : 'Stable / Baseline'}\\n\\n`;

    text += `MANDATORY CONSULTATIONS & ESCALATIONS:\\n`;
    if (state.triggers.wocRNConsult) {
      text += `[!] MANDATORY CONSULT: WOC-RN (Wound, Ostomy, Continence Specialist)\\n`;
      text += `    └ Trigger: ${state.isWorsened ? 'Wound WORSENER' : 'Stage 3+ / Full-Thickness / HAPI criteria met'}\\n`;
    }
    if (state.triggers.podiatryVascular) {
      text += `[!] PRIMARY CONSULT: Podiatry / Vascular Surgery\\n`;
      text += `    └ Trigger: Full-thickness / Stage 3+ lesion on lower extremity/foot.\\n`;
    }
    if (state.triggers.surgicalEval) {
      text += `[!] SURGICAL EVALUATION: Sharp / Surgical Debridement Consult\\n`;
      text += `    └ Trigger: Presence of slough or necrotic eschar.\\n`;
    }
    if (state.triggers.nutritionConsult) {
      text += `[!] MANDATORY CONSULT: Registered Dietitian / Clinical Nutrition\\n`;
      text += `    └ Trigger: Full-thickness loss / Stage 3+ / Multiple Stage 2 PIs.\\n`;
    }
    if (!Object.values(state.triggers).some(Boolean)) {
      text += `- Routine protocol care. No specialty escalations currently triggered.\\n`;
    }

    text += `\\nRECOMMENDED TOPICAL CARE PLAN (App Formulary Match):\\n`;
    text += `1. Cleanse: Normal Saline or non-cytotoxic wound cleanser.\\n`;
    text += `2. Periwound Protection: Apply No-Sting Barrier Film to surrounding skin.\\n`;
    if (state.hasSloughOrEschar) {
      text += `3. Debridement / Moisture: Apply enzymatic debrider per order OR moisture-retentive gel. (Note: If tissue is severely dry, consider adding neutral wound gel to activate enzymatic process).\\n`;
    }
    if (state.hasTunnelingOrUndermining || isFullThickness) {
      text += `4. Cavity / Primary Layer: Gently pack cavity/tunneling with Hydrofiber or Calcium Alginate rope. (Gelling fiber action preferred over WTD when viable tissue is present).\\n`;
      text += `5. Cover Layer: Non-Adherent Contact Layer + Secondary Non-Bordered Gauze OR Bordered Silicone Foam.\\n`;
    } else {
      text += `3. Primary Cover: Bordered Silicone Foam. (Note: Helps reduce friction and shear, NOT pressure).\\n`;
    }

    text += `\\nMECHANICAL INTERVENTIONS & DME (Non-Dressing Interventions):\\n`;
  }
}
```

```

    if (isFootOrAnkle) {
      text += `• OFF-LOADING: Strict offloading of feet required at all times.\n`;
      text += `• DME DEVICE: Apply heel suspension boot / offloading boot per order. (Offloading boots are DME, NEVER classified as a
dressing).\n`;
    } else {
      text += `• PRESSURE REDISTRIBUTION: Q2H turning schedule, sacral offloading wedge, and pressure-redistributing mattress
overlay.\n`;
    }
    if (state.isIncontinent) {
      text += `• MOISTURE BARRIER: Apply zinc-based barrier paste or liquid barrier film to perineal area after each incontinence
exposure.\n`;
    }

    text += `\n===== \n`;
    text += `Disclaimer: Decision support only. Validate against facility protocol.\n`;
    text += `"There are a thousand ways to treat a wound, but what does your patient really need?"\n`;
    text += `===== `;

    return text;
  };

const copyToClipboard = () => {
  navigator.clipboard.writeText(generateNote());
  setCopied(true);
  setTimeout(() => setCopied(false), 2000);
};

return (
  <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
    { /* Header */ }
    <div className="mb-6 flex items-center justify-between">
      <div>
        <h2 className="text-2xl font-bold text-slate-900">Recommended Care Plan & Orders</h2>
        <p className="text-sm text-slate-600">Standardized clinical documentation and specialty routing.</p>
      </div>
      <span className="text-xs font-semibold px-3 py-1 bg-green-100 text-green-800 rounded-full">
        Step 4 of 4 (Complete)
      </span>
    </div>

    { /* Copy Note Box */ }
    <div className="mb-6 bg-slate-900 rounded-lg p-4 border border-slate-800">
      <div className="flex justify-between items-center mb-3">
        <span className="text-xs font-bold text-slate-400 uppercase tracking-wide">
          EHR Ready Documentation Note
        </span>
        <button
          onClick={copyToClipboard}
          className="px-3 py-1.5 bg-blue-600 hover:bg-blue-500 text-white rounded text-xs font-semibold transition"
        >
          {copied ? '✓ Copied to Clipboard' : 'Copy Note'}
        </button>
      </div>
      <pre className="text-xs text-slate-300 font-mono whitespace-pre-wrap overflow-x-auto max-h-96 p-3 bg-slate-950 rounded border
border-slate-800">
        {generateNote()}
      </pre>
    </div>

    { /* Key Safety Reminders Banner */ }
    <div className="p-4 bg-amber-50 border border-amber-200 rounded-lg mb-8 text-xs text-amber-900 space-y-1">
      <p className="font-bold">Clinical Safety Reminders:</p>
      <p>• <strong>Silicone Foams:</strong> Help reduce friction and shear, <strong>NOT pressure</strong>.</p>
      <p>• <strong>Offloading Boots:</strong> Durable Medical Equipment (DME), <strong>NEVER a dressing option</strong>.</p>
      <p>• <strong>Gelling Fibers (Hydrofibers):</strong> Preferred over Wet-to-Dry (WTD) when clean viable tissue is present and
packing/wicking is required.</p>
    </div>

    { /* Action Controls */ }
    <div className="flex justify-between pt-4 border-t border-slate-200">
      <button
        type="button"
        onClick={onBack}
        className="px-5 py-2.5 rounded-lg text-sm font-medium border border-slate-300 text-slate-700 hover:bg-slate-50"
      >
        ← Edit Assessment
      </button>
      <button
        type="button"

```

```
        onClick={onReset}
        className="px-6 py-2.5 rounded-lg text-sm font-semibold bg-slate-900 text-white hover:bg-slate-800 shadow-md"
      >
        Start New Patient Assessment 
      </button>
    </div>
  </div>
);
};
```

08 app/components/Navbar.tsx

```
'use client';

import React from 'react';
import { NavigationModule, UserRole } from '../../../types/wound';

interface NavbarProps {
  activeModule: NavigationModule;
  setActiveModule: (module: NavigationModule) => void;
  role: UserRole;
  resetAssessment: () => void;
}

export const Navbar: React.FC<NavbarProps> = ({
  activeModule,
  setActiveModule,
  role,
  resetAssessment,
}) => {
  const navItems: { id: NavigationModule; label: string }[] = [
    { id: 'ASSESSMENT', label: 'Triage & Assessment' },
    { id: 'FORMULARY', label: 'Dressing Formulary' },
    { id: 'DEBRIDEMENT', label: 'Debridement Guide' },
    { id: 'GLOSSARY', label: 'Clinical Glossary' },
    { id: 'CASES', label: 'Case Studies' },
  ];

  return (
    <header className="w-full shadow-md">
      { /* Tier 1: Royal Blue Top Branding Banner */ }
      <div className="bg-blue-950 text-white px-4 py-3 sm:px-8 flex flex-col sm:flex-row items-center justify-between border-b border-blue-800">
        <div className="flex items-center space-x-3">
          <span className="bg-blue-600 text-white font-extrabold text-xs px-2.5 py-1 rounded tracking-wide shadow-sm">
            POCKET WOC
          </span>
          <h1 className="text-lg font-bold tracking-tight text-slate-100">
            Clinical Decision Support System
          </h1>
        </div>

        <div className="mt-2 sm:mt-0 flex items-center space-x-4">
          <a
            href="https://www.skintegritypartners.com"
            target="_blank"
            rel="noopener noreferrer"
            className="text-xs font-semibold text-blue-200 hover:text-white transition-colors bg-blue-900/60 px-3 py-1.5 rounded-full border border-blue-700/50 flex items-center gap-1.5"
          >
            <span>Skintegrity Partners LLC</span>
            <span className="text-[10px]">></span>
          </a>
          {role && (
            <button
              onClick={resetAssessment}
              className="text-xs font-medium bg-red-800/80 hover:bg-red-700 text-white px-2.5 py-1 rounded transition-colors"
            >
              Reset Session
            </button>
          )}
        </div>
      </div>

      { /* Tier 2: Lower Module Navigation Banner */ }
      <nav className="bg-slate-800 text-slate-200 px-4 py-2 sm:px-8 overflow-x-auto">
        <div className="flex items-center justify-start sm:justify-center space-x-2 sm:space-x-4 min-w-max">
          {navItems.map((item) => {
            const isActive = activeModule === item.id;
            return (
              <button
                key={item.id}
                onClick={() => setActiveModule(item.id)}
                className={`px-3.5 py-1.5 rounded-md text-xs font-semibold transition-all ${
                  isActive
                    ? 'bg-blue-600 text-white shadow-sm'
                    : 'text-slate-300 hover:bg-slate-700 hover:text-white'
                }`}
              >
                {item.label}
              </button>
            );
          })}
        </div>
      </nav>
    </header>
  );
};
```

```
        }}  
      >  
        {item.label}  
      </button>  
    );  
  }}  
</div>  
</nav>  
</header>  
)  
};
```

09 app/components/RoleSelection.tsx

```
'use client';

import React from 'react';
import { UserRole, WoundState } from '../../../types/wound';

interface RoleSelectionProps {
  state: WoundState;
  updateState: (updates: Partial<WoundState>) => void;
  onNext: () => void;
}

export const RoleSelection: React.FC<RoleSelectionProps> = ({
  state,
  updateState,
  onNext,
}) => {
  const handleRoleSelect = (role: UserRole) => {
    updateState({ role });
  };

  const handlePaceSelect = (paceMode: 'FAST' | 'DETAILED') => {
    updateState({ paceMode });
  };

  const canProceed = state.role !== null;

  return (
    <div className="max-w-3xl mx-auto py-8 px-4 sm:px-6">
      {/* Intro Header */}
      <div className="text-center mb-6">
        <h2 className="text-2xl font-bold text-slate-900 sm:text-3xl">
          Clinical Decision Support
        </h2>
        <p className="mt-2 text-sm text-slate-600">
          Select your clinical role and assessment parameters to load the appropriate care pathway.
        </p>
      </div>

      {/* Skintegrity Partners Quote Block */}
      <div className="bg-blue-50 border-l-4 border-blue-600 p-4 rounded-r-lg mb-8 shadow-sm">
        <p className="text-xs italic text-blue-900 leading-relaxed">
          &ldquo;Standardizing wound triage and dressing selection empowers bedside clinicians, drives evidence-based practice, and optimizes tissue healing outcomes across care settings.&rdquo;
        </p>
        <p className="mt-2 text-[11px] font-bold text-blue-950 uppercase tracking-wider">
          – Skintegrity Partners LLC
        </p>
      </div>

      {/* Role Selection & Pace Mode Card */}
      <div className="bg-white rounded-xl shadow-sm border border-slate-200 p-6 space-y-6">
        <div>
          <label className="block text-xs font-bold uppercase tracking-wider text-slate-700 mb-3">
            1. Select Clinical Role
          </label>
          <div className="grid grid-cols-2 gap-4">
            {[ 'RN', 'PROVIDER' ] as UserRole[] }.map((r) => (
              <button
                key={r}
                type="button"
                onClick={() => handleRoleSelect(r)}
                className={`py-3 px-4 rounded-lg border text-sm font-semibold transition-all ${
                  state.role === r
                    ? 'border-blue-600 bg-blue-50 text-blue-900 ring-2 ring-blue-500/20'
                    : 'border-slate-200 bg-slate-50 text-slate-700 hover:bg-slate-100'
                }`
              >
                {r === 'RN' ? 'Registered Nurse (RN)' : 'Provider (MD/DO/NP/PA)'}
              </button>
            ))
          </div>
        </div>

        <div>
          <label className="block text-xs font-bold uppercase tracking-wider text-slate-700 mb-3">
```

```

    2. Triage Mode
  </label>
  <div className="grid grid-cols-2 gap-4">
    {[
      { id: 'FAST', label: 'Fast Track (Rapid Triage)' },
      { id: 'DETAILED', label: 'Detailed Assessment' },
    ]}.map((mode) => (
      <button
        key={mode.id}
        type="button"
        onClick={() => handlePaceSelect(mode.id as 'FAST' | 'DETAILED')}
        className={`py-2.5 px-3 rounded-lg border text-xs font-medium transition-all ${
          state.paceMode === mode.id
            ? 'border-blue-600 bg-blue-50 text-blue-900 font-semibold'
            : 'border-slate-200 bg-slate-50 text-slate-600 hover:bg-slate-100'
          }`}
      >
        {mode.label}
      </button>
    ))}
  </div>
</div>

{/* Continue Button */}
<div className="pt-4 border-t border-slate-100 flex justify-end">
  <button
    onClick={onNext}
    disabled={!canProceed}
    className={`px-6 py-2.5 rounded-lg text-sm font-bold text-white transition-all ${
      canProceed
        ? 'bg-blue-600 hover:bg-blue-700 shadow-md cursor-pointer'
        : 'bg-slate-300 cursor-not-allowed'
      }`}
  >
    Proceed to Location Selection →
  </button>
</div>
</div>
</div>
);
};

```

10 app/components/BodyAvatar.tsx

```
'use client';

import React from 'react';

type AnatomicalZone = string;

interface BodyAvatarProps {
  selectedZone: AnatomicalZone | null;
  onSelectZone: (zone: AnatomicalZone) => void;
  showTypical: boolean;
  showAtypical: boolean;
}

export const BodyAvatar: React.FC<BodyAvatarProps> = ({
  selectedZone,
  onSelectZone,
  showTypical,
  showAtypical,
}) => {
  const zones = [
    { id: 'Occiput (Back of Head)', label: 'Occiput', cx: 100, cy: 38, isTypical: false },
    { id: 'Sacrum', label: 'Sacrum', cx: 100, cy: 195, isTypical: true },
    { id: 'Coccyx', label: 'Coccyx', cx: 100, cy: 212, isTypical: true },
    { id: 'Ischial Tuberosity', label: 'Ischium', cx: 88, cy: 228, isTypical: true },
    { id: 'Greater Trochanter', label: 'Trochanter', cx: 68, cy: 220, isTypical: true },
    { id: 'Malleolus (Ankle)', label: 'Malleolus', cx: 122, cy: 345, isTypical: false },
    { id: 'Heel (Calcaneus)', label: 'Heel', cx: 78, cy: 368, isTypical: true },
    { id: 'Foot / Toes', label: 'Foot / Toes', cx: 122, cy: 375, isTypical: false },
  ];

  return (
    <div className="flex flex-col items-center bg-slate-950 p-5 rounded-2xl border border-blue-900/60 shadow-2xl text-white w-full relative overflow-hidden">
      {/* HUD Background Visual Effect */}
      <div className="absolute inset-0 bg-[radial-gradient(circle_at_center,var(--tw-gradient-stops))] from-blue-900/20 via-slate-950 to-slate-950 pointer-events-none" />

      {/* Header */}
      <div className="mb-3 text-center z-10">
        <div className="inline-flex items-center gap-1.5 px-2.5 py-0.5 rounded-full bg-blue-900/40 border border-blue-500/30 mb-1">
          <span className="w-1.5 h-1.5 rounded-full bg-cyan-400 animate-ping" />
          <span className="text-[10px] font-mono tracking-wider text-cyan-300 uppercase">
            LDA VECTOR MAP
          </span>
        </div>
        <h4 className="text-sm font-bold text-slate-100">Clinical Body Avatar</h4>
      </div>

      {/* Vector Container */}
      <div className="relative w-64 h-[410px] bg-slate-900/80 rounded-xl p-3 border border-slate-800 flex justify-center items-center backdrop-blur-sm z-10">
        <svg viewBox="0 0 200 400" className="w-full h-full drop-shadow-[0_0_12px_rgba(59,130,246,0.15)]">
          <defs>
            {/* Tech Grid Pattern */}
            <pattern id="grid" width="20" height="20" patternUnits="userSpaceOnUse">
              <path d="M 20 0 L 0 0 0 20" fill="none" stroke="rgba(51, 65, 85, 0.3)" strokeWidth="0.5" />
            </pattern>
            {/* Gradient Fill for Body Outline */}
            <linearGradient id="bodyGrad" x1="0%" y1="0%" x2="100%" y2="100%">
              <stop offset="0%" stopColor="#1e293b" stopOpacity="0.9" />
              <stop offset="50%" stopColor="#0f172a" stopOpacity="0.95" />
              <stop offset="100%" stopColor="#1e1b4b" stopOpacity="0.9" />
            </linearGradient>
          </defs>

          {/* Background Grid */}
          <rect width="200" height="400" fill="url(#grid)" />

          {/* HUD Reticle Overlay */}
          <circle cx="100" cy="200" r="170" fill="none" stroke="rgba(56, 189, 248, 0.08)" strokeWidth="1" strokeDasharray="4 4" />
          <circle cx="100" cy="200" r="120" fill="none" stroke="rgba(56, 189, 248, 0.05)" strokeWidth="1" />

          {/* High-Detail Vector Body Silhouette */}
          <path
            d="

```

```

M 100 18
C 107 18, 114 24, 114 34
C 114 42, 109 48, 107 51
C 114 54, 126 58, 134 65
C 142 72, 147 88, 149 105
C 151 122, 148 145, 145 160
C 143 170, 139 175, 136 172
C 134 165, 137 140, 137 125
C 133 135, 131 160, 129 180
C 127 195, 126 210, 126 225
C 125 245, 128 275, 127 305
C 126 325, 125 345, 125 360
C 125 372, 128 382, 132 385
C 126 388, 116 388, 114 382
C 113 372, 116 350, 115 330
C 114 300, 108 260, 103 235
C 101 225, 100 220, 100 220
C 100 220, 99 225, 97 235
C 92 260, 86 300, 85 330
C 84 350, 87 372, 86 382
C 84 388, 74 388, 68 385
C 72 382, 75 372, 75 360
C 75 345, 74 325, 73 305
C 72 275, 75 245, 74 225
C 74 210, 73 195, 71 180
C 69 160, 67 135, 63 125
C 63 140, 66 165, 64 172
C 61 175, 57 170, 55 160
C 52 145, 49 122, 51 105
C 53 88, 58 72, 66 65
C 74 58, 86 54, 93 51
C 91 48, 86 42, 86 34
C 86 24, 93 18, 100 18 Z
"
fill="url(#bodyGrad)"
stroke="#38bdf8"
strokeWidth="1.2"
strokeLinejoin="round"
/>

{/* Anatomical Landmark Zones */}
{zones.map((zone) => {
  const isSelected = selectedZone === zone.id;
  const highlight = (zone.isTypical && showTypical) || (!zone.isTypical && showAtypical);

  return (
    <g key={zone.id} onClick={() => onSelectZone(zone.id)} className="cursor-pointer group">
      {/* Prominence Halo */}
      {highlight && (
        <circle
          cx={zone.cx}
          cy={zone.cy}
          r="15"
          className={` ${
            zone.isTypical
              ? 'fill-amber-500/30 stroke-amber-400'
              : 'fill-purple-500/30 stroke-purple-400'
          } animate-pulse`}
          strokeWidth="1"
          strokeDasharray="2 2"
        />
      )}
    </g>
  )
})

{/* Reticle Target Line for Selection */}
{isSelected && (
  <g className="animate-spin" style={{ transformOrigin: `${zone.cx}px ${zone.cy}px`, animationDuration: '8s' }}>
    <circle cx={zone.cx} cy={zone.cy} r="11" fill="none" stroke="#ef4444" strokeWidth="1" strokeDasharray="3 3" />
  </g>
)}

{/* Core LDA Pin Dot */}
<circle
  cx={zone.cx}
  cy={zone.cy}
  r={isSelected ? '6' : '4'}
  className={
    isSelected
      ? 'fill-red-500 stroke-white stroke-2'
      : 'fill-cyan-400 group-hover:fill-white group-hover:scale-125 transition-all'
  }
/
```

```

    }
  />

  { /* Callout Label */}
  <text
    x={zone.cx > 100 ? zone.cx + 12 : zone.cx - 12}
    y={zone.cy + 3}
    textAnchor={zone.cx > 100 ? 'start' : 'end'}
    className={`text-[9px] font-mono tracking-tight pointer-events-none transition-all ${
      isSelected
        ? 'fill-red-400 font-bold text-[10px]'
        : 'fill-slate-400 group-hover:fill-cyan-300'
      }}`
    >
    {zone.label}
  </text>
</g>
);
}}
</svg>
</div>
</div>
);
};

```

11 app/components/InteractiveBody3D.tsx

```
'use client';

import React, { useRef, useEffect, Suspense } from 'react';
import { Canvas, useFrame, useThree } from '@react-three/fiber';
import { OrbitControls, useFBX, Html } from '@react-three/drei';
import * as THREE from 'three';

interface TargetZone {
  id: string;
  name: string;
  targetPos: [number, number, number];
  cameraPos: [number, number, number];
  tagOffset: [number, number, number];
}

const TARGET_ZONES: Record<string, TargetZone> = {
  'Malleolus (Ankle)': {
    id: 'Malleolus (Ankle)',
    name: 'Malleolus (Ankle)',
    targetPos: [0.18, -1.05, 0.05],
    cameraPos: [0.18, -0.9, 0.75],
    tagOffset: [0.18, 0.08, 0],
  },
  'Heel (Calcaneus)': {
    id: 'Heel (Calcaneus)',
    name: 'Heel (Calcaneus)',
    targetPos: [-0.18, -1.15, -0.05],
    cameraPos: [-0.18, -1.0, 0.75],
    tagOffset: [-0.18, 0.08, 0],
  },
  'Sacrum': {
    id: 'Sacrum',
    name: 'Sacrum',
    targetPos: [0, -0.05, -0.15],
    cameraPos: [0, -0.05, -0.85],
    tagOffset: [0.22, 0.1, 0],
  },
  'Greater Trochanter': {
    id: 'Greater Trochanter',
    name: 'Greater Trochanter',
    targetPos: [0.3, -0.15, 0],
    cameraPos: [0.75, -0.15, 0.5],
    tagOffset: [0.18, 0.1, 0],
  },
  'Ischial Tuberosity': {
    id: 'Ischial Tuberosity',
    name: 'Ischial Tuberosity',
    targetPos: [-0.12, -0.25, -0.1],
    cameraPos: [-0.12, -0.25, -0.8],
    tagOffset: [-0.18, 0.1, 0],
  },
  'Foot / Toes': {
    id: 'Foot / Toes',
    name: 'Foot / Toes',
    targetPos: [0.18, -1.18, 0.18],
    cameraPos: [0.18, -1.0, 0.75],
    tagOffset: [0.18, 0.08, 0],
  },
};

function MannequinMesh({ selectedZone }: { selectedZone: string | null }) {
  const fbx = useFBX('/Models/Ch36_nonPBR.fbx');
  const activeZone = selectedZone ? TARGET_ZONES[selectedZone] : null;

  return (
    <group>
      { /* Base Mannequin Mesh */ }
      <primitive>
        object={fbx}
        scale={0.012}
        position={[0, -1.2, 0]}
      />

      { /* Transparent Glass Highlight & Callout Pin */ }
      {activeZone && (
```

```

<group position={activeZone.targetPos}>
  { /* Outer Transparent Glow Ring */}
  <mesh>
    <sphereGeometry args={[0.075, 24, 24]} />
    <meshPhysicalMaterial
      color="#38bdf8"
      transparent
      opacity={0.25}
      roughness={0.1}
      transmission={0.8}
      thickness={0.2}
      depthWrite={false}
    />
  </mesh>

  { /* Inner Wireframe Core Marker */}
  <mesh>
    <sphereGeometry args={[0.045, 16, 16]} />
    <meshStandardMaterial
      color="#06b6d4"
      transparent
      opacity={0.5}
      wireframe
      depthWrite={false}
    />
  </mesh>

  { /* Center Point */}
  <mesh>
    <sphereGeometry args={[0.015, 12, 12]} />
    <meshBasicMaterial color="#67e8f9" transparent opacity={0.85} />
  </mesh>

  { /* Callout Badge pinned to target */}
  <Html position={activeZone.tagOffset} center distanceFactor={2.5}>
    <div className="bg-slate-900/85 border border-cyan-400/70 text-cyan-200 font-mono text-[10px] font-bold px-2.5 py-1 rounded-
md shadow-lg whitespace-nowrap backdrop-blur-md flex items-center gap-1.5 pointer-events-none">
      <span className="w-1.5 h-1.5 rounded-full bg-cyan-400 animate-ping" />
      {activeZone.name}
    </div>
  </Html>
</group>
)}
</group>
);
}

// Camera Rig allowing free manual orbit after auto-transition
function CameraRig({ selectedZone }: { selectedZone: string | null }) {
  const { camera } = useThree();
  const controlsRef = useRef<any>(null);
  const activeZoneRef = useRef<string | null>(null);
  const isTransitioning = useRef<boolean>(false);

  // Trigger camera move only when selectedZone actually changes
  useEffect(() => {
    if (selectedZone !== activeZoneRef.current) {
      activeZoneRef.current = selectedZone;
      isTransitioning.current = true;
    }
  }, [selectedZone]);

  useFrame(() => {
    if (!isTransitioning.current || !controlsRef.current) return;

    let targetCam: THREE.Vector3;
    let targetLookAt: THREE.Vector3;

    if (selectedZone && TARGET_ZONES[selectedZone]) {
      const zone = TARGET_ZONES[selectedZone];
      targetCam = new THREE.Vector3(...zone.cameraPos);
      targetLookAt = new THREE.Vector3(...zone.targetPos);
    } else {
      targetCam = new THREE.Vector3(0, -0.1, 3.0);
      targetLookAt = new THREE.Vector3(0, -0.1, 0);
    }

    // Smoothly lerp position and controls target
    camera.position.lerp(targetCam, 0.07);
  });
}

```

```

controlsRef.current.target.lerp(targetLookAt, 0.07);
controlsRef.current.update();

// End transition lock once camera is near destination so user has full orbit control
if (camera.position.distanceTo(targetCam) < 0.02) {
  isTransitioning.current = false;
}
});

return (
  <OrbitControls
    ref={controlsRef}
    enablePan={true}
    enableZoom={true}
    enableRotate={true}
    minDistance={0.3}
    maxDistance={4.5}
  />
);
}

export default function InteractiveBody3D({
  selectedZone,
}): {
  selectedZone: string | null;
}) {
  return (
    <div className="w-full h-[520px] bg-slate-800 rounded-xl overflow-hidden relative border border-slate-700 shadow-xl">
      { /* Dynamic HUD Header */ }
      <div className="absolute top-3 left-3 z-10 flex items-center gap-2 bg-slate-900/80 border border-slate-700 px-3 py-1.5 rounded-full backdrop-blur-md">
        <span className={`w-2 h-2 rounded-full ${selectedZone ? 'bg-cyan-400 animate-pulse' : 'bg-emerald-400'} `} />
        <span className="text-[11px] font-mono font-bold text-slate-200 uppercase tracking-wider">
          {selectedZone ? `BONY PROMINENCE: ${selectedZone}` : `3D MANNEQUIN (FULL VIEW)`}
        </span>
      </div>

      <Canvas camera={{ position: [0, -0.1, 3.0], fov: 45 }}>
        <ambientLight intensity={1.1} />
        <directionalLight position={[5, 8, 5]} intensity={1.5} />
        <directionalLight position={[-5, -5, -3]} intensity={0.6} />

        <Suspense fallback={<Html center><div className="text-cyan-400 font-mono text-xs">Loading Model...</div></Html>}>
          <MannequinMesh selectedZone={selectedZone} />
        </Suspense>

        <CameraRig selectedZone={selectedZone} />
      </Canvas>
    </div>
  );
}

```

12 app/components/MaleAnatomyModel.tsx

```
import { Suspense } from 'react';
import { Html } from '@react-three/drei';

export function MaleAnatomyModel({ selectedZone }: { selectedZone: string }) {
  return (
    <group>
      <mesh>
        <boxGeometry args=[[1, 1.5, 0.35]] />
        <meshStandardMaterial color={selectedZone ? '#d1d5db' : '#9ca3af'} />
      </mesh>
    </group>
  );
}

// Inside your Canvas scene:
export default function MaleAnatomyScene({ selectedZone }: { selectedZone: string }) {
  return (
    <Suspense fallback={<Html><div className="text-white">Loading 3D Model...</div></Html>}>
      <MaleAnatomyModel selectedZone={selectedZone} />
    </Suspense>
  );
}
```

13 app/components/ModuleViews.tsx

```
'use client';

import React, { useState } from 'react';

// =====
// 1. DRESSING FORMULARY VIEW
// =====
export const FormularyView: React.FC = () => {
  return (
    <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
      <div className="mb-6">
        <h2 className="text-2xl font-bold text-slate-900">Generic Dressing Formulary</h2>
        <p className="text-sm text-slate-600">Product-agnostic categories, mechanisms, and application rules.</p>
      </div>

      <div className="space-y-4">
        { /* Bordered Silicone Foams */ }
        <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
          <div className="flex items-center justify-between mb-2">
            <h3 className="text-base font-bold text-slate-900">Bordered Silicone Foams</h3>
            <span className="text-xs font-semibold px-2.5 py-1 bg-blue-100 text-blue-800 rounded">Moderate-Heavy Exudate</span>
          </div>
          <p className="text-xs text-slate-700 mb-2">
            <strong>Mechanism:</strong> Multi-layer absorbency. <strong>Helps reduce friction and shear, NOT pressure.</strong>
          </p>
          <p className="text-xs text-slate-600 mb-1"><strong>Indications:</strong> Shallow to full-thickness wounds; sacral/heel friction prevention.</p>
          <p className="text-xs text-slate-500"><strong>Change Frequency:</strong> Q3D-Q5D and PRN for strikethrough or soilage.</p>
        </div>

        { /* Hydrofibers & Calcium Alginates */ }
        <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
          <div className="flex items-center justify-between mb-2">
            <h3 className="text-base font-bold text-slate-900">Hydrofibers & Calcium Alginates</h3>
            <span className="text-xs font-semibold px-2.5 py-1 bg-amber-100 text-amber-800 rounded">Cavity / Heavy Exudate</span>
          </div>
          <p className="text-xs text-slate-700 mb-2">
            <strong>Mechanism:</strong> Soft fibers that turn into a smooth gel when absorbing fluid, locking in moisture and bacteria.
          </p>
          <p className="text-xs text-slate-600 mb-1">
            <strong>Clinical Preference:</strong> Gelling action is preferred over wet-to-dry (WTD) dressings when clean, viable tissue is present and cavity packing/wicking is the main goal.
          </p>
          <p className="text-xs text-slate-500"><strong>Safety Alert:</strong> Do NOT use on dry wounds or dry eschar. Do not combine silver variants with enzymatic debriders.</p>
        </div>

        { /* Contact Layers & Barrier Films */ }
        <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
          <div className="flex items-center justify-between mb-2">
            <h3 className="text-base font-bold text-slate-900">Contact Layers & Periwound Barrier Films</h3>
            <span className="text-xs font-semibold px-2.5 py-1 bg-emerald-100 text-emerald-800 rounded">Tissue / Periwound Protection</span>
          </div>
          <p className="text-xs text-slate-700 mb-2">
            <strong>Mechanism:</strong> Porous non-adherent mesh sheets and no-sting polymer films protecting periwound skin against maceration and adhesive stripping.
          </p>
          <p className="text-xs text-slate-600 mb-1"><strong>Indications:</strong> Fragile granulation tissue, skin tears, and high incontinence exposure areas.</p>
        </div>
      </div>
    </div>
  );
};

// =====
// 2. DEBRIDEMENT GUIDE VIEW
// =====
export const DebridementView: React.FC = () => {
  return (
    <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
      <div className="mb-6">
        <h2 className="text-2xl font-bold text-slate-900">Debridement Scope & Matrix</h2>
        <p className="text-sm text-slate-600">Differentiating RN scope from provider and surgical interventions.</p>
      </div>
    </div>
  );
};
```

```

</div>

<div className="grid grid-cols-1 md:grid-cols-2 gap-4">
  { /* Autolytic */ }
  <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
    <span className="px-2 py-0.5 text-[10px] font-bold bg-blue-100 text-blue-800 rounded uppercase">RN Independent Scope</span>
    <h3 className="text-base font-bold text-slate-900 mt-2 mb-1">1. Autolytic Debridement</h3>
    <p className="text-xs text-slate-600 mb-2">Uses body's own natural enzymes under moisture-retentive dressings (hydrogels,
medical honey, films).</p>
    <p className="text-xs text-slate-500"><strong>Best For:</strong> Uninfected wounds with dry eschar or slough where speed is not
urgent.</p>
  </div>

  { /* Enzymatic */ }
  <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
    <span className="px-2 py-0.5 text-[10px] font-bold bg-purple-100 text-purple-800 rounded uppercase">Provider Order
Required</span>
    <h3 className="text-base font-bold text-slate-900 mt-2 mb-1">2. Enzymatic Debrider</h3>
    <p className="text-xs text-slate-600 mb-2">Prescribed topical ointment digesting collagen bonds attaching dead tissue to the
wound bed.</p>
    <p className="text-xs text-amber-900 bg-amber-50 p-2 rounded text-[11px]">
      <strong>Clinical Note:</strong> If target tissue is severely dry, consider adding moisture via neutral wound gel or saline to
activate the enzyme.
    </p>
  </div>

  { /* Conservative Sharp */ }
  <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm md:col-span-2">
    <span className="px-2 py-0.5 text-[10px] font-bold bg-red-100 text-red-800 rounded uppercase">Provider / Surgical Scope</span>
    <h3 className="text-base font-bold text-slate-900 mt-2 mb-1">3. Surgical & Sharp Debridement</h3>
    <p className="text-xs text-slate-600 mb-2">Fast bedside or operating room removal of non-viable tissue using scalpel, curette,
or scissors.</p>
    <p className="text-xs text-slate-700"><strong>Urgent Escalation:</strong> Immediate Podiatry, Vascular, or General Surgery
consult required for rapidly spreading infection, gross necrosis, or wet gangrene.</p>
  </div>
</div>
);
};

// =====
// 3. CLINICAL GLOSSARY VIEW
// =====
export const GlossaryView: React.FC = () => {
  const [search, setSearch] = useState('');

  const terms = [
    { term: 'Slough', def: 'Yellow, tan, or stringy devitalized tissue. Presence automatically classifies wound as Full-Thickness loss.' },
    { term: 'Eschar', def: 'Dark (black or brown) thick, leathery dead tissue attached to a wound bed.' },
    { term: 'DTPI (Deep Tissue Pressure Injury)', def: 'Intact or non-intact skin presenting as localized persistent non-blanchable deep red, maroon, or purple discoloration.' },
    { term: 'Tunneling / Sinus Tract', def: 'Narrow channel extending from wound bed in any direction into deeper tissue.' },
    { term: 'Undermining', def: 'Tissue destruction extending under intact skin along wound margins.' },
    { term: 'Incontinence-Associated Dermatitis (IAD)', def: 'Moisture damage from urine or stool causing diffuse shiny inflammation without distinct margins.' },
  ];

  const filtered = terms.filter(t => t.term.toLowerCase().includes(search.toLowerCase()) || t.def.toLowerCase().includes(search.toLowerCase()));

  return (
    <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
      <div className="mb-6">
        <h2 className="text-2xl font-bold text-slate-900">Clinical Glossary</h2>
        <p className="text-sm text-slate-600 mb-4">Characteristic-based definitions and visual cues.</p>
        <input
          type="text"
          placeholder="Search clinical terms (e.g., Slough, DTPI)..."
          value={search}
          onChange={(e) => setSearch(e.target.value)}
          className="w-full p-2.5 rounded-lg border border-slate-300 text-xs focus:ring-2 focus:ring-blue-500 focus:outline-none"
        />
      </div>

      <div className="space-y-3">
        {filtered.map((item, idx) => (
          <div key={idx} className="p-4 bg-white rounded-lg border border-slate-200">
            <h3 className="text-sm font-bold text-blue-900">{item.term}</h3>

```

```

        <p className="text-xs text-slate-600 mt-1">{item.def}</p>
      </div>
    )}}
  </div>
</div>
);
};

// =====
// 4. CASE STUDIES VIEW
// =====
export const CaseStudiesView: React.FC = () => {
  return (
    <div className="max-w-4xl mx-auto py-8 px-4 sm:px-6">
      <div className="mb-6">
        <h2 className="text-2xl font-bold text-slate-900">Case Studies & Practice Scenarios</h2>
        <p className="text-sm text-slate-600">Bedside decision exercises demonstrating tool workflow.</p>
      </div>

      <div className="p-4 bg-slate-900 text-white rounded-lg mb-6 text-xs space-y-2">
        <p className="font-bold text-blue-400">Clinical Disclaimer & Escalation Protocol:</p>
        <p>Pocket WOC is a decision support tool—not a substitute for careful assessment. Always escalate and reference your facility
Wound Care specialist when needed.</p>
        <p className="italic text-slate-300 pt-1">"There are a thousand ways to treat a wound, but what does your patient really need?"
</p>
      </div>

      <div className="space-y-4">
        <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
          <div className="flex justify-between items-center mb-2">
            <h3 className="text-sm font-bold text-slate-900">Case 1: Sacral Pressure Injury (Stage 3 vs. Unstageable)</h3>
            <span className="text-[10px] font-bold bg-green-100 text-green-800 px-2 py-0.5 rounded">UNLOCKED</span>
          </div>
          <p className="text-xs text-slate-600 mb-2">>3cm x 2cm crater with yellow stringy slough covering upper right quadrant.</p>
          <div className="text-xs text-slate-500 bg-slate-50 p-3 rounded space-y-1">
            <p>• <strong>Slough Detected:</strong> Confirms full-thickness loss (Stage 3 / Unstageable equivalent).</p>
            <p>• <strong>Triggers:</strong> WOC-RN Consult + Dietitian/Nutrition Consult.</p>
            <p>• <strong>Formulary Plan:</strong> Hydrofiber packing + Bordered Silicone Foam cover.</p>
          </div>
        </div>

        <div className="p-5 bg-white rounded-lg border border-slate-200 shadow-sm">
          <div className="flex justify-between items-center mb-2">
            <h3 className="text-sm font-bold text-slate-900">Case 2: Heel DTPI vs. Ecchymosis</h3>
            <span className="text-[10px] font-bold bg-green-100 text-green-800 px-2 py-0.5 rounded">UNLOCKED</span>
          </div>
          <p className="text-xs text-slate-600 mb-2">>4cm x 4cm dark purple intact area on heel following surgery.</p>
          <div className="text-xs text-slate-500 bg-slate-50 p-3 rounded space-y-1">
            <p>• <strong>System Classification:</strong> Deep Tissue Pressure Injury (DTPI).</p>
            <p>• <strong>Triggers:</strong> Primary Podiatry / Vascular Consult + WOC-RN Consult.</p>
            <p>• <strong>Mandatory Rule:</strong> Strict offloading of feet required. Apply heel boot (DME, NOT dressing).</p>
          </div>
        </div>
      </div>
    </div>
  );
};

```